

NeuroNimbus: A Cloud-Powered AI System for Memory Reconstruction

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Abstract—NeuroNimbus is a cloud-assisted intelligent system that is aimed at assisting individuals with cognitive impairments to perform digital memory reconstruction processes. Memory disorders make it difficult for individuals to remember people and events; hence, they are forced to depend on caregivers and manual memory storage mechanisms. To address these issues, the proposed system uses a dual interface approach that includes a caregiver interface and a patient interface. Users will be able to upload images with accompanying captions using a web portal. Natural Language Processing (NLP) is used to analyze the uploaded captions to extract contextual keywords. TF-IDF is used to represent features and perform cosine similarity analysis to ensure consistency of memories stored by detecting redundant memories. Memories are organized under person and event categories depending on the assigned labels. NeuroNimbus will be able to provide an organized digital memory support system by integrating semantic processing, redundancy control, and security mechanisms with role-based access control. Experimental validation demonstrated an overall functional accuracy of 90.2%, indicating stable and reliable system performance in structured memory reconstruction tasks.

Index Terms—Digital Memory Reconstruction, Natural Language Processing (NLP), TF-IDF, Cognitive Support Systems, Cloud Computing, Role-Based Access Control, Memory Organization, Duplicate Detection, Healthcare Informatics.

I. INTRODUCTION

Memory is one of the most important aspects of human life. It enables a person to remember past experiences, identify known people, and preserve a personal identity. But certain neurological disorders like Alzheimer's and dementia significantly impair a person's ability to remember events, people, and everyday details [1], [2]. As the disease advances, people find it difficult to remember significant memories and become more reliant on others for help. Although tremendous progress has been made in various fields like communication, healthcare, and education, memory-aiding technology remains largely manual, such as notebooks, printed photo albums, and oral instructions [3]. These traditional techniques can become cumbersome and mentally taxing for both patients and caregivers in the long run.

Individuals suffering from memory impairment also have to deal with issues such as a lack of digital literacy, the complexity of memory interfaces, and a need for caregivers to handle personal data [4]. Moreover, many existing digital memory storage systems lack intelligent organization. Pictures and descriptions are often unorganized, leading to repeated entries, ambiguous descriptions, and the inability to retrieve significant information. Without the use of automated keyword identification, similarity analysis, and contextual grouping, memory storage becomes disorganized and mentally taxing, rather than memory-aiding [5], [6].

To address these problems, this paper presents NeuroNimbus: A Cloud-Based AI System for Memory Reconstruction, which is a web-based application that employs Natural Language Processing (NLP) tasks and cloud computing resources [7]. The application allows users to upload images and corresponding descriptive text. The text is then processed using NLP tasks to extract relevant keywords and information, making it easier to organize and structure memories into person-related and event-related groups based on assigned labels and contextual metadata. This makes it easier to organize and manage memories without needing to have in-depth technical knowledge. Simple approaches such as TF-IDF Keyword Extraction and Cosine Similarity are used to automatically detect repeated or similar entries of memories, ensuring that the memory database is well-organized and free from redundancy [8], [9]. The application also incorporates secure cloud storage with role-based access control, featuring separate interfaces for patients and caregivers and ensuring synchronized and scalable data management [10].

Existing memory support applications are generally limited to manual recording or individual functionality without using intelligent automation and understanding [3], [5]. Most applications lack the capability for semantic processing and do not support automatic memory categorization and redundancy management, making them less efficient in a well-organized healthcare setting.

Through the integration of NLP-based contextual analysis and secure cloud services, NeuroNimbus provides a user-friendly and scalable solution for digital memory management. The system focuses on usability, lightweight processing, and organized storage, making it ideal for a healthcare setting where computational power and technical know-how might be constrained.

The main contributions of this paper are as follows:

- (i) A cloud-supported system for structured digital memory reconstruction;
- (ii) A keyword extraction and similarity-based categorization system using NLP for grouping memories into significant categories;
- (iii) A redundancy detection system for maintaining repository consistency, which is lightweight;
- (iv) A dual-interface and role-based system design to minimize cognitive overload for patients as well as caregivers.

The rest of this paper is organized as follows: Section II provides a review of the existing literature on AI-based healthcare support systems and digital memory management systems. Section III discusses the architecture and approach of the proposed system. Section IV provides the implementation details and validation of the proposed system. Section V provides implications and discussions of the proposed system, and conclusions and future work are presented in Section VI.

II. LITERATURE REVIEW

Conventional memory support systems for senior citizens have traditionally relied on manual methods such as journals, physical photo albums, and audio recordings, where caregivers play a pivotal role in managing personal memories. Although these methods are emotionally significant, they lack digital organization and automation. Consequently, these systems have been inefficient, error-prone, susceptible to data loss, and cognitively taxing for both patients and caregivers. Furthermore, patients suffering from cognitive impairments have been known to experience challenges in using complex digital interfaces, thereby creating a pressing need for simplified, intelligent, and automated memory management systems enabled by AI, NLP, and cloud computing.

The Digital Memory Book (2015) [1] focused on personalized photo-based reminiscence therapy to improve patient engagement. Nevertheless, the project was highly dependent on manual data entry and did not include automated keyword extraction, similarity identification, or cloud integration. Memory Lane (SoftwareX, 2025) [2] brought forth the use of intent classification and CNN-based face recognition to help Alzheimer’s patients recognize familiar faces using voice commands. Although highly technologically advanced, the project was primarily centered on face recognition and not text-based memory reconstruction. Additionally, the project’s dependency on computationally intensive models could potentially hinder its use in resource-limited healthcare environments.

The Extractive Text Summarization Review (IEEE Access, 2025) [3] demonstrated the effectiveness of NLP methods like TF-IDF and transformer models for text-related tasks.

Nevertheless, the research mainly covered general text summarization tasks and did not target personalized digital memory management. In the same manner, the Multimodal Brain Decoding study (Brain Research Bulletin, 2023) [4] contributed to AI-based language decoding from brain signals but mainly focused on neurobiological analysis. The research on Deep Learning in the Cloud (Neurocomputing) [5] examined the architecture of scalable AI systems but mainly targeted infrastructure optimization rather than memory reconstruction applications for end-users.

In general, the current body of research focuses primarily on individual aspects like face recognition, voice control, text summarization, neural decoding, or cloud scalability. Few systems, however, attempt to combine automated semantic reconstruction, contextual keyword extraction, redundancy removal, systematic categorization, and secure cloud synchronization in a single, patient-oriented approach. This is a substantial research gap in the area of digital memory assistance systems.

To address this research gap, the proposed NeuroNimbus framework integrates TF-IDF keyword extraction, cosine similarity-based duplicate identification, person- and event-level categorization, and cloud storage in a scalable web-based system. By focusing on light semantic processing, role-based access control, and systematic organization, NeuroNimbus hopes to offer a feasible solution for cognitive assistance and long-term digital memory management.

TABLE I
SUMMARY OF EXISTING RESEARCH AND IDENTIFIED GAPS

Research Area	Key References
Manual Memory Systems	[1]
Face Recognition & Voice Assistance	[2]
NLP Processing (TF-IDF)	[3]
Neuroscience AI	[4]
Cloud AI Deployment	[5]
Similarity Detection (TF-IDF)	[3], [5]
Integrated Memory Framework (NeuroNimbus)	Proposed Work

The structured review above highlights that although a lot of progress has been made in AI-based healthcare technologies, an integrated, lightweight, and patient-centric digital memory reconstruction framework is still under-explored. NeuroNimbus fills this research gap by integrating semantic processing, context organization, redundancy elimination, and secure cloud storage into a single deployable system.

III. PROPOSED METHOD

This section introduces the architecture and flow of the proposed NeuroNimbus framework for structured digital memory reconstruction. The framework is intended to convert unstructured image caption data into a structured, searchable, and contextually informed memory repository through light-weight Natural Language Processing and secure cloud hosting. Unlike conventional human-operated memory storage solutions, the

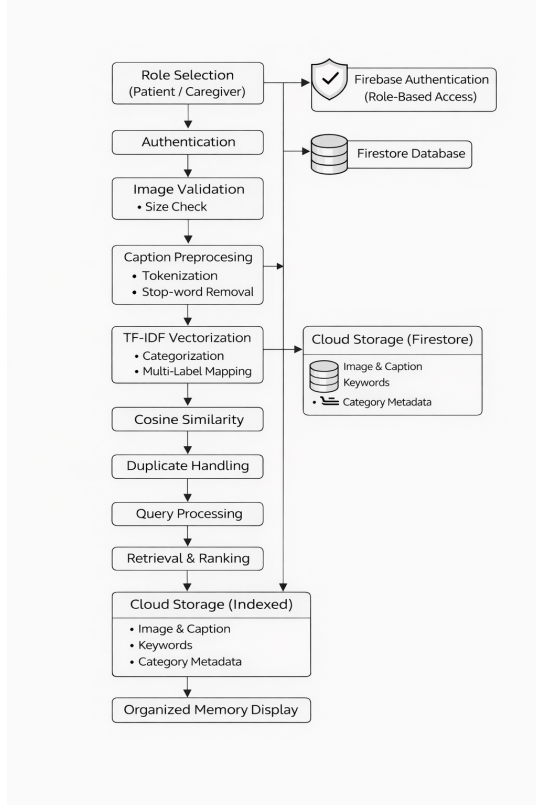


Fig. 1. Architecture

proposed framework combines semantic analysis, redundancy management, and role based access control in a single solution.

As shown in Fig. 1, the framework has a structured workflow that includes role-based authentication, memory upload, text preprocessing, keyword extraction, similarity identification, structured categorization, secure cloud storage, and retrieval.

The entire workflow of the NeuroNimbus application is designed in a way such that the raw user input is gradually refined and transformed into a well-organized and searchable memory bank. As the caregiver adds a memory entry by uploading the data, the application validates the information and matches it with the appropriate patient profile. The caption is refined and converted into a numerical form using TF-IDF. Similarity checks are also conducted before adding the data to the database in order to avoid any redundancy. Keywords are also indexed in the data for easier search and browsing.

A. Role-Based Architecture and Identity Management

NeuroNimbus uses a two-role architecture that includes a caregiver interface and a patient interface. The caregiver is tasked with creating and managing memory repositories. When patients register in the system, they are given a system-generated unique identifier to prevent data duplication. The system has uniqueness constraints on patient names and identifiers to ensure structured storage and prevent overlapping entries.

The patient interface is used to grant controlled access

to view categorized memories, special events, and caregiver-uploaded notes. Authentication is done through cloud-based role validation to ensure that each patient can only access their memory repository.

B. Memory Input and Validation

The process of memory reconstruction is initiated when a caregiver uploads an image with a caption and a corresponding person label. The system also allows additional contextual information, such as reminders or special events like date and title. To ensure that the content is clear and consistent, the system uses validation constraints, which include minimum image size requirements to avoid poor-quality images. Every memory entry is associated with a particular patient repository based on the unique identifier created during the registration process. The uploaded memories are then submitted to the preprocessing and analysis stage.

C. Text Preprocessing

Each caption is normalized to eliminate linguistic noise and facilitate semantic extraction. The preprocessing step involves lowercasing, tokenization, and stop-word removal. This processing step is necessary to prepare the text content for feature extraction while retaining significant contextual terms.

The memory dataset can be represented as:

$$M = \{(I_i, C_i)\}_{i=1}^N \quad (1)$$

where:

- I_i represents the uploaded image,
- C_i represents the corresponding caption,
- N is the total number of memory entries.

D. NLP-Based Keyword Extraction and Categorization

To represent textual information numerically, TF-IDF vectorization is applied.

The term frequency (TF) of a term t in caption C_i is defined as:

$$TF(t, C_i) = \frac{f(t, C_i)}{\sum_k f(k, C_i)} \quad (2)$$

The inverse document frequency (IDF) is given by:

$$IDF(t) = \log \left(\frac{N}{df(t)} \right) \quad (3)$$

The TF-IDF score is computed as:

$$TFIDF(t, C_i) = TF(t, C_i) \times IDF(t) \quad (4)$$

Each caption is thus represented as a feature vector:

$$V_i = [w_1, w_2, \dots, w_m] \quad (5)$$

where m denotes the vocabulary size and w_k represents the TF-IDF weight of term k .

The extracted keywords are used to improve contextual organization and retrieval within the repository. Memories in the proposed system are organized in person-centered groups and event-centered groups based on labels and associated metadata supplied during the memory upload.

For person-centered memories, the system will group memories based on the individual label associated with the memory by the caregiver, facilitating structured access within the People section. Event-centered memories will be organized based on event metadata such as event name and date, facilitating retrieval within dedicated event categories.

The memory grouping mechanism may be conceptually depicted as follows:

$$F : V_i \rightarrow G \quad (6)$$

where:

- G represents the structured memory group determined by assigned labels and event metadata,
- V_i provides contextual enhancement for search and similarity-based organization.

E. Similarity Detection and Redundancy Control

To ensure consistency in the repository, the cosine similarity measure is used between the memory vectors as follows:

$$Sim(V_i, V_j) = \frac{V_i \cdot V_j}{\|V_i\| \|V_j\|} \quad (7)$$

If:

$$Sim(V_i, V_j) > \tau \quad (8)$$

where τ is the predefined similarity threshold, the vectors are marked as duplicates or near-duplicates.

If similar images are uploaded with different contextual labels, the system retains a single image instance and maps multiple labels to it. This multi-association mapping helps in avoiding redundancy while maintaining semantic associations. This similarity-driven redundancy control ensures efficient cloud utilization and prevents repository clutter.

F. Cloud Storage and Secure Synchronization

The processed memory components, such as images, captions, extracted keywords, and category information, are stored in the cloud infrastructure.

The cloud layer supports scalable storage, real-time synchronization, role-based authentication, and logical separation of patient repositories. The cloud-based architecture supports horizontal scalability by separating authentication, storage, and indexing services. As the number of memory entries increases, indexing mechanisms ensure that retrieval performance remains efficient. The efficient use of an index ensures that retrieval is quick even with an increase in entries. Logical separation of patient repositories ensures stable concurrent access without data interference.

The patient repositories are indexed based on the unique identifier assigned during the registration process for secure storage.

G. Search and Retrieval Mechanism

The search and retrieval mechanism supports keyword search, person-grouping, and event-filtering. When a search query is entered, the system searches the TF-IDF vectors and corresponding metadata for matching query keywords to retrieve the relevant memories.

The retrieval function can be expressed as:

$$R(Q) = \{M_i \mid relevance(V_i, Q) > \delta\} \quad (9)$$

where:

- Q represents the user query,
- V_i represents the TF-IDF vector of memory entry M_i ,
- δ is the predefined relevance threshold.

The result is shown in an organized memory display with images, captions, context tags, and categorized groups. The organized search and retrieval system is more convenient and user-friendly for people with memory disorders.

H. Theoretical Justification and Design Considerations

Although transformer-based deep learning architectures yield more complex contextual embeddings, they demand large amounts of labeled data and heavy computational resources. However, the proposed TF-IDF-based approach provides interpretable, efficient semantic representation with lower complexity and faster processing time.

The computational complexity of TF-IDF vectorization can be approximated as:

$$O(Nm) \quad (10)$$

where:

- N represents the total number of memory entries,
- m represents the vocabulary size.

This makes the approach feasible for implementation in a healthcare setting with limited infrastructure.

By emphasizing the importance of interpretability, efficiency, and organized structure, the proposed system is theoretically feasible without the need for resource-hungry deep learning models.

IV. EXPERIMENTS AND RESULTS

A. Dataset Description

In order to validate the efficiency and usability of the proposed NeuroNimbus framework, experiments were carried out using a dataset consisting of 125 memory entries. These memory entries were collected through the developed web-based application. These memory entries were created within the application environment and represent real scenarios, including person-related and event-related memories.

Each memory entry is represented by an image-caption pair along with its contextual information. More specifically, the memory entry is represented by an associated image file, a caption representing the content of the image, a person-related label representing the subject, and optionally contextual information such as notes and special event information such as the title and the date of the event.

Out of the total entries, approximately 60% of the memory entries were for person-centered memory, and the rest 40% of the memory entries were for event-centered memory. This helped in ensuring a balanced validation of the categorization module.

TABLE II
SUMMARY OF THE EXPERIMENTAL DATASET

Dataset Attribute	Value
Total Memory Entries	125
Person-centered Memories	75 (60%)
Event-centered Memories	50 (40%)
Data Type	Image + Caption
Additional Metadata	Person Label, Event Title, Date, Notes

The characteristics of the dataset used for experimental evaluation are summarized in Table II.

It was also found that the textual captions of the images varied in their lengths and styles of writing. Some of the captions were short and descriptive in nature, while others contained detailed narrative descriptions. This helped in creating a realistic environment in order to test the NLP-based keyword extraction module for its effectiveness in handling different kinds of natural languages and their usages, which included incomplete information and minor spelling mistakes.

The dataset had both structured and unstructured data components. The structured data components included person tags, timestamps, and event identifiers, while the unstructured data component was primarily in the form of free-form captions. This ensured that the testing of the system’s functionality in terms of its ability to transform heterogeneous raw inputs into a structured, organized, and searchable memory repository was comprehensive.

The dataset was used to validate the functionality of the three main functional modules of the system, namely keyword extraction from the captions, structured grouping into person-based and event-based groups based on assigned labels and metadata, and similarity-driven duplicate detection with multi-label association. Given the nature of the proposed framework, which is based on light-weight statistical and rule-based NLP techniques and not supervised model training, the dataset is appropriate for the testing of the system’s functionality.

B. Experimental Setup

The system was tested on three main functional components: keyword extraction, memory categorization, and duplicate detection.

The system was tested for functionality through validation of system output against expected results. For each component, the system output was validated for:

- Correct identification of context-specific keywords,
- Correct grouping into person- and event-related categories,
- Correct identification and elimination of duplicate entries.

The performance of the system was evaluated by comparing the system output with the expected output, which was manually verified. Accuracy of each module was determined by calculating the ratio of the number of correctly processed memory entries to the total number of tested entries. This evaluation approach provides a practical measure of functional

reliability for each module within the overall memory reconstruction pipeline. Note that, as the proposed framework is based on statistical NLP and rule-based similarity measures, there was no need to train the model, as in the case of traditional approaches.

System-level testing also included:

- Validation of role-based access control,
- Verification of cloud synchronization stability,
- Latency testing under multiple user interactions,
- Isolation testing of patient repositories.

C. Results and Performance Analysis

The experimental results indicate stable and consistent system behavior for all modules.

The Memory Categorization module performed well in grouping captions with contextual markers. The Duplicate Detection module successfully avoided redundant storage of similar images using cosine similarity thresholds and maintained multi-label associations when necessary.

The Keyword Extraction module was slightly sensitive to language variations, including incomplete captions and minor spelling inconsistencies. However, normalization and stop-word filtering improved the quality of keyword extraction.

TABLE III
MODULE-WISE PERFORMANCE EVALUATION

Module	Accuracy (%)
Keyword Extraction	87.6
Memory Categorization	90.8
Duplicate Detection	92.3

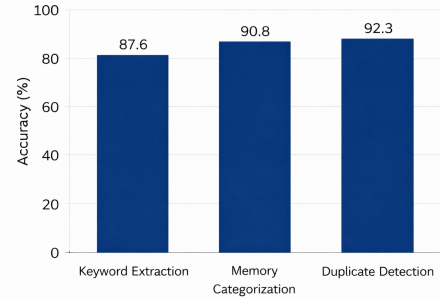


Fig. 2. Performance comparison of NeuroNimbus modules based on evaluation accuracy

The performance of individual modules is summarized in Table III, and the comparative accuracy is visually illustrated in Fig. 2. The results show that the duplicate detection module recorded the highest accuracy as a result of the deterministic nature of cosine similarity thresholding, and the performance of the keyword extraction module was slightly affected by natural language captions. The analysis shows that the categorization and duplicate detection modules had higher

stability than the keyword extraction module, mainly because of the deterministic nature of grouping logic and similarity thresholding.

Cloud synchronization remained stable during all experiments. There were no instances of cross-patient data leakage and authentication errors. Multi-user experiments showed acceptable latency and stable repository updates.

User-level experiments also validated the effectiveness of person-label, event category, and keyword-based memory grouping for improved accessibility and easier search. The memory repository organization improved usability in cognitive support applications.

In addition to that, the overall functional accuracy of the proposed framework was calculated by considering the average accuracy of all three modules of interest. As per the results obtained and presented in Table III, it was found that the system was able to achieve an accuracy of 90.2% in total.

The experimental results demonstrate that NeuroNimbus successfully integrates NLP-based semantic analysis, redundancy control using similarity measures, structured categorization, and secure cloud storage for a functional digital memory reconstruction system.

V. CONCLUSION AND FUTURE WORK

This paper introduced NeuroNimbus, a cloud-assisted intelligent framework for structured digital memory reconstruction. The proposed system incorporates TF-IDF-based keyword extraction, cosine similarity-driven redundancy detection, structured person- and event-based grouping based on assigned labels and metadata, and role-based cloud storage in a single framework.

The experimental results demonstrated that the system exhibited stable functional characteristics for all modules of the system. The keyword extraction module was able to extract the appropriate descriptors from the user-generated captions, and the memory categorization module enabled structured grouping of memories into meaningful person- and event-based categories. The redundancy detection module was able to reduce redundant storage using the similarity threshold, thus maintaining consistency in the repository.

Rather than relying on computationally complex deep learning models, the framework emphasizes interpretability, lightweight semantic processing, and deployability. This makes the system applicable in the healthcare setting, where computational complexity and resource management are of concern. The organized storage and retrieval functionality reduces caregiver workload and enhances accessibility for people with memory impairments.

Although the current implementation demonstrates feasibility and stability, there are still some avenues for future improvements. The addition of transformer-based contextual embeddings can increase semantic complexity and context understanding. The addition of speech-to-text functionality can increase accessibility for people with limited typing ability. Further enhancement of memory organization with contextual tagging using temporal or location metadata can also be

explored. Multilingual support and integration with healthcare information systems can also increase the applicability of the framework in the healthcare setting.

Practically speaking, the lightweight architecture of the framework makes it suitable for real-world health environments such as small clinics, assisted living facilities, and home-care environments. The absence of large training datasets and computationally intensive deep learning models makes the framework suitable for environments with low technical resources. Therefore, the NeuroNimbus framework presents a suitable solution for long-term cognitive support.

In summary, NeuroNimbus provides a scalable and patient-centric foundation for AI-assisted digital memory management. The system illustrates the real-world viability of integrating light NLP methodologies with secure cloud infrastructure to facilitate long-term cognitive care.

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REFERENCES

- [1] M. Azam, S. Khalid, S. Almutairi, H. A. Khattak, A. Namoun, A. Ali, and H. S. M. Bilal, "Current trends and advances in extractive text summarization: A comprehensive review," *IEEE Access*, vol. 13, pp. 1–25, 2025.
- [2] R. Che, J. Li, and H. Wang, "Transforming reminiscence therapy: Designing an adaptive VR system based on older adults' emotional connections and narrative patterns," in *Proc. IEEE Int. Symp. Mixed and Augmented Reality Adjunct (ISMAR-Adjunct)*, 2025, pp. 815–816.
- [3] J. S., L. S., T. Ragupathi, K. Hariharan, R. Sanjana, and D. G. Arora, "Personalized dementia prediction using machine learning techniques," in *Proc. Int. Conf. Power, Control and Computing Technologies (ICPCT)*, 2025, pp. 101–105.
- [4] D.-G. Lee and S.-B. Lee, "Diagnosis of Alzheimer's disease and frontotemporal dementia from electroencephalography signals," *IEEE Transactions on Neural Systems and Rehabilitation Engineering*, vol. 33, pp. 2160–2169, 2025.
- [5] Z.-J. Lin, Y.-J. Chen, P.-C. Kuo, L. Huang, C.-J. Hu, and C.-Y. Chen, "Dementia assessment using Mandarin speech with an attention-based speech recognition encoder," in *Proc. IEEE Int. Conf. Acoustics, Speech and Signal Processing (ICASSP)*, 2024, pp. 12461–12465.
- [6] A. Balamurali, H. K. R., and H. S., "Early detection of dementia: Machine learning for predictive analysis," in *Proc. Int. Conf. Distributed Computing and Internet of Things (IDCIoT)*, 2024, pp. 973–980.
- [7] C. I. Gonzalez, "Towards a general type-2 fuzzy reasoning approach to modeling modifiable dementia risk factors," in *Proc. Int. Conf. Soft Computing and Machine Intelligence (ISCMI)*, 2024, pp. 205–210.

- [8] S. Routray, D. Samir, G. Pushya, and J. Srivastava, "Early detection of dementia using a large language model-powered chatbot," in *Proc. Int. Conf. Power, Distributed Computing and Grid Computing (PDGC)*, 2024, pp. 719–724.
- [9] Y. B. Fadhel, M. Rinchi, and K. Al-Haddad, "Smart device for monitoring persons with Alzheimer disease," in *Proc. Int. Conf. Artificial Intelligence and Green Energy (ICAIGE)*, 2024, pp. 1–6.
- [10] R. K. Ranjan and D. Rai, "A statistical study of Lewy body and other types of dementia," in *Proc. Int. Conf. Intelligent Systems and Pattern Recognition (INSPECT)*, 2024, pp. 1–6.
- [11] K. Poonam, R. Guha, and P. P. Chakrabarti, "Artificial intelligence based hierarchical classification of frontotemporal dementia," in *Proc. IEEE Engineering in Medicine and Biology Society Conf. (EMBC)*, 2024, pp. 1–4.
- [12] M. M. Murshid, J. H. Mamun, and M. R. Rafee, "A robust technique for prediction of dementia using ensemble voting classifier," in *Proc. Int. Conf. Computing, Communication and Networking Technologies (ICCCNT)*, 2024, pp. 1–5.
- [13] K. Y. Chan, B. Abu-Salih, R. Qaddoura, A. M. Al-Zoubi, V. Palade, D.-S. Pham, J. Del Ser, and K. Muhammad, "Deep neural networks in the cloud: Review, applications, challenges and research directions," *Neurocomputing*, vol. 545, p. 126327, 2023.
- [14] Y. Zhao, Y. Chen, K. Cheng, and W. Huang, "Artificial intelligence based multimodal language decoding from brain activity: A review," *Brain Research Bulletin*, vol. 201, p. 110713, 2023.
- [15] M. S. Kabir, J. Khanom, M. A. Bhuiyan, Z. N. Tumpa, S. F. Rabby, and S. Bilgaiyan, "Early detection of dementia disease using machine learning approach," in *Proc. Int. Conf. Computer Communication and Informatics (ICCCI)*, 2023, pp. 1–6.
- [16] P. Reisinho, R. Raposo, and N. Zagalo, "A systematic literature review of virtual reality-based reminiscence therapy for people with cognitive impairment or dementia," in *Proc. Int. Conf. Information Systems and Technologies (IMET)*, 2022, pp. 1–8.
- [17] A. Islam et al., "A predictive analysis for early signs of dementia," in *Proc. Int. Conf. Information Technology (ICIT)*, 2022, pp. 1–5.
- [18] M. R. Lima et al., "Conversational affective social robots for ageing and dementia support," *IEEE Transactions on Cognitive and Developmental Systems*, vol. 14, no. 4, pp. 1378–1397, 2022.
- [19] P.-F. Wu, S.-W. Yen, and K.-Y. Fan, "Investigating the influence through EEG: Using technology to introduce cognitive training for patients with mild dementia," in *Proc. Int. Conf. Biomedical Engineering and Informatics (ECBIOS)*, 2021, pp. 38–40.
- [20] A. Bannon, A. Rapeaux, and T. G. Constandinou, "Tiresias: A low-cost networked UWB radar system for in-home monitoring of dementia patients," in *Proc. IEEE Engineering in Medicine and Biology Society Conf. (EMBC)*, 2021, pp. 7068–7072.